but there is a decided objection due to the arrangements not being the most economical. With the motors connected permanently in parallel there is considerable power lost in the resistance. Economy can be obtained by connecting the motors in combinations. Less resistance is required and there is less loss in the resistance.

The amount of energy loss is shown by Fig. 2. The figure is based on an electric locomotive equipped with four motors to operate on a third rail at 650 volts. It will require 1,000 amps per motor to start up this locomotive coupled to a certain train, and accelerating it up to a certain speed (where all the resistance is cut out) in 45 seconds.

With each motor taking 1,000 amps at 650 volts the power from the line is (4 x 1,000) x 650 = 2,600 K. W. This number of kilowatts is taken during the straight line acceleration, and is represented by the line *ab*. Forty-five seconds has elapsed so that the energy used is shown graphically by the area *o a b c*, and is 2,600 K. W. x 45 sec. = 117,000 K. W. seconds or

$$\frac{17{,}000}{3{,}600 \text{ (seconds per hr.)}} = 32.5 \text{ K. W. H.}$$

The energy lost in the resistance depends on the product of the current and the volts lost in the resistance. The volts lost is the difference between the line volts and the motor volts. A certain voltage is necessary to start the motor and is assumed as 10% of the line voltage so that in our present problem it is 65 volts. The kilowatts taken by the motors is then

$$\frac{65 \times 4{,}000}{1{,}000} = 260 \text{ kw. and}$$

this value locates point X. After 45 seconds all resistance is out, so that the line *xb* represents the energy lost in the resistance. The area below the line represents the useful energy absorbed by the motor. As 10% of the line voltage is used at the start, 90% will be lost in the resistance at start. Since *xb* is a straight line the average volts will be one-half of 90, or 45% of 650, which is 292.5 volts. The energy loss is therefore 292.5 x 4,000 x 45 ÷ by 3,600 x 1,000 = 14.6 K.W.H.

# Details of Parts of the Pacific Type Locomotive as Shown in Our Chart, No. 12

**Spring Suspension of Engine Frame.**

The illustration of the parts of the chart of the Pacific locomotive, presented herewith, shows the arrangement of the spring suspension of the main frame. The arrangements of the suspension of the front and rear trucks were shown in the issues of January and March, 1920, respectively, and will only be indicated in this place.

In the case of the main frame the parts lying below the spring suspension are the wheels, axles, driving boxes and driving spring saddles and stirrups.

Taking up the whole structure as shown in this illustration, we first have the frame. This is usually a steel casting at the present time, though formerly it was invariably an iron forging. It is formed of an upper rail (1) and lower rail (2), with pedestal legs (3 and 4), a front slab section (5), to which the cylinders are bolted, and a rear slab section (6) upon which the firebox of the boiler rests and which, in turn, is carried by an equalizer connecting the main spring suspension with that of the rear truck.

For a Pacific locomotive there are three sets of pedestals for the front (7), main (8) and rear (9) driving boxes, respectively. These driving boxes are made up of two pieces, the box proper (7, 8 and 9), and a driving box brass (10). These are sometimes called crown brasses. The box itself may be of cast iron or cast steel, and the fit for the brasses cut out, usually on a slotting machine, to a semicircular form and the brass pressed into place by hydraulic pressure. The lower portion of the box is fitted with a cellar (11), which is merely a box of cast iron having its upper portion curved to come close to the axle (13). It is fitted with oil and waste packed in so as to be in contact with the axle journal and is held up in place by the driving box axle pin (12).

The frame pedestals are braced and prevented from spreading under the thrusts of the axle boxes by the pedestal tie bars (14). These bars are first fitted to bear against the pedestal tie bar lugs (15) and then bolted to the frame by the pedestal tie bar bolts (16).

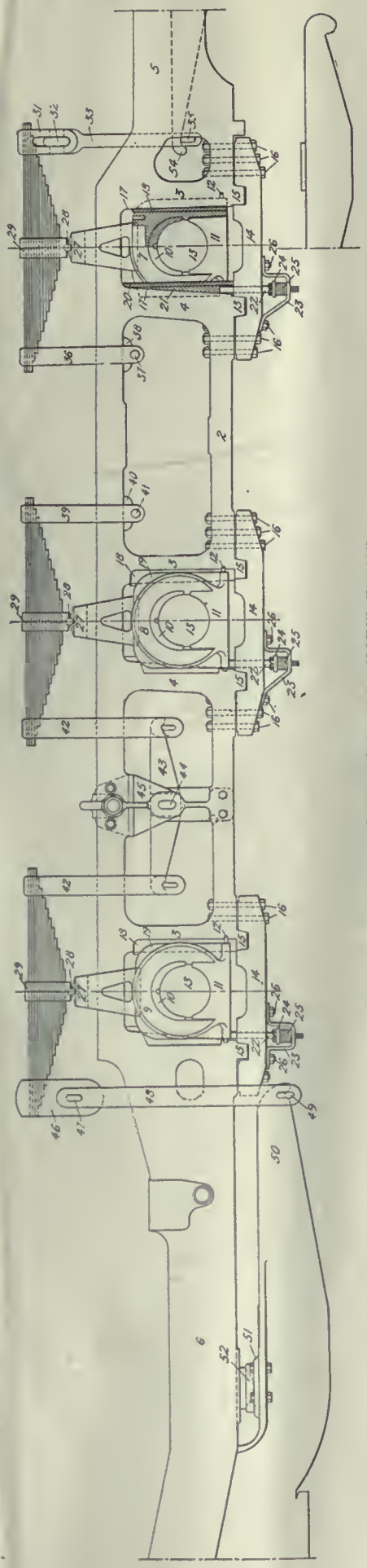
The inner faces of the pedestal legs are protected against the wear that would be occasioned by the up and down movement of the driving boxes by a set of shoes and wedges. It will be noticed that the inner faces of the pedestal legs (17) are not vertical but are inclined so that the distance between them is greater at the bottom than at the top. To the front leg a shoe (18) is rigidly fastened. This shoe is of a U shape in horizontal section, the inner side and bottom fitting against the leg of the pedestal and the outside planed to go between the front flanges (19) of the driving box. The rear pedestal leg is protected by a similar shoe (20), called an adjusting wedge, which bears against the driving box, but which is made adjustable so as to take up any wear that may occur by a floating wedge (21) placed between it and the face of the rear pedestal leg. There are a number of designs of this shoe and wedge. Usually they are combined in a single casting, but in the one shown they are in two pieces and are known as the Franklin automatic adjustable driving box wedge parts and are automatically adjustable. That is to say the wedge is automatically raised so as to push the shoe against the driving box and thus take up any wear that may occur.

This is accomplished in this case by means of the adjustable wedge bolt (22) which is attached to the wedge and extends down through the pedestal tie bar and the Franklin adjustable driving box wedge spring bracket (23). The bolt is threaded at its lower end and carries a spring cap (24). This is used to compress the spring (25). This spring is thus under compression and exerts a constant upward pressure against the bolt and, through it on the adjustable wedge, tending to force the latter upward between the face of the pedestal leg and the shoe, thus taking up all lost motion of the axle boxes due to wear. The spring bracket is held to the pedestal tie bar by the studs (26).

Turning now to the top of the driving box, it will be found to be fitted with a pocket on either side of the frame for the reception of the two legs of the spring stirrup or saddle (27). This is usually a steel casting and spans the frame. On its top is the driving spring roll (28) on which the spring band (29) rests, and on which it is free to rock with the movement of the semi-elliptic driving spring (30), as it is compressed or relaxed under the influence of the working of the load.

We now come to the spring suspension proper.

Starting at the front, the end of the spring of the leading driving wheel is spanned by a short spring suspension link (31), which carries the end of the cross equalizer (32). This equalizer runs across the engine from side to side, and is carried at each end in the manner here described. This cross equalizer is a heavy forging or steel casting, and has a driver and truck equalizer hanger (33) sus-



DETAILS OF SPRING SUSPENSION OF ENGINE FRAME.

pended from its center. The lower end spans the back end of the driver and truck equalizer (34), which it supports by means of the driver and truck equalizer hanger key (35). This equalizer has a bearing on a pin beneath the cylinder saddle, while its forward end rests on the truck.

The back end of the front spring carries the spring hanger (36), which spans the frame and carries it by means of the spring hanger pin (37), which carries the frame through the spring hanger spring seat (38) attached to the bottom of the top rail of the frame.

Thus there is an equalization or a distribution of the weight of the front end of the engine, between the front driving wheel springs and those of the front truck.

In the case of the spring suspension of the rear end of the engine we find the front end of the main driving spring attached to the frame by the means of a hanger (39) and pin (40) and seat (41), which are the exact duplicates of the hanger, pin and seat (36, 37 and 38).

From the back end of the main driving spring and the front end of the rear driving spring, there are two similar spring and equalizer hangers (42) which carry the two ends of the equalizer (43), respectively. This equalizer carries a portion of the frame between the main and rear driving wheels by means of the equalizer key (44), which has a bearing in the equalizer fulcrum casting (45), which is bolted to and between the top and bottom rails of the frame.

From back end of the rear driver spring, a combination of link (46), link pin (47), spring hanger (48) and spring hanger key (49), support the forward end of the trailing truck and driver equalizer (50). This equalizer forms a portion of the trailing truck as illustrated in our issue for March, 1920, and, as there described, carries the rear end of the engine frame through the fulcrum socket bracket (51) and the fulcrum ball (52).

Thus, as the equalization of the spring support at the front end of the engine causes the front driver and front truck springs to act in unison, so that at the rear it produces a similar unity of action between the main and rear driving springs and those of the trailing truck.

At the back end, however, the springs on the two sides of the engine are allowed to act independently of each other, while, at the front they are tied together through the cross equalizer (32). This, in a final analysis, resolves itself into a three-point support which is characteristic of American locomotive construction, and is the feature which gives to the type its extraordinary flexibility of action and freedom from liability to serious fracture.

The following is the list of parts.condom from liability to serious fracture or breakage.

List of Parts.

1. Upper rail of frame.
2. Lower rail of frame.
3. Front leg of pedestal.
4. Back leg of pedestal.
5. Front slab section.
6. Rear slab section.
7. Front driving box.
8. Main driving box.
9. Rear driving box.
10. Driving box brass.
11. Driving box cellar.
12. Driving box cellar pin.
13. Driving axle.
14. Pedestal tie bar.
15. Pedestal tie bar lugs.
16. Pedestal tie bar bolts.
17. Inner face of pedestal legs.
18. Front pedestal shoe.
19. Driving box front flange.
20. Franklin automatic adjusting wedge.
21. Franklin automatic floating wedge.
22. Franklin automatic driving box wedge bolt.
23. Franklin automatic adjustable wedge spring bracket.
24. Franklin automatic adjustable wedge spring cap.
25. Franklin automatic adjustable wedge spring.
26. Franklin automatic adjustable wedge spring bracket stud.
27. Driving spring stirrup.
28. Driving spring roller.
29. Driving spring band.
30. Driving spring.
31. Spring suspension link.
32. Cross equalizer.
33. Driver and truck equalizer hanger.
34. Driver and truck equalizer.
35. Driver and truck equalizer hanger key.
36. Front driving spring hanger.
37. Spring hanger pin.
38. Spring hanger pin seat.
39. Spring driving spring front hanger.
40. Main driving spring front hanger pin.
41. Main driving spring front hanger pin seat.
42. Driving spring and equalizer hanger.
43. Spring equalizer.
44. Spring equalizer key.
45. Equalizer fulcrum casting.
46. Rear driver spring link.
47. Rear driver spring link pin.
48. Rear driver spring back hanger.
49. Rear driver spring back hanger key.
50. Trailing truck and driver equalizer.
51. Equalizer fulcrum socket bracket.
52. Equalizer fulcrum ball.

# Air Brake Department

## Questions and Answers

### Locomotive Air Brake Inspection

(*Continued from page* 370, *Dec.*, 1919.)

1031. Q.—What are some of the bad habits a man can get into in the manipulation of brake valve handles?

A.—Allowing the handle to remain in running position when charging a freight train; coupling to a charged passenger train with the valve handle on lap position; too long a time in release position; allowing the valve handle to stop on lap position before a service application; moving the handle from lap to running or holding position without first moving to release position; moving the handle away from running position at any time except in actual cases of brakes sticking or for applying brakes; using partial emergency position for a service stop, and constantly using the independent brake valve in release position when the automatic valve is in running position.

1032. Q.—What do you conclude from watching a man make these movements to release position with the independent valve while the automatic valve is in running position?

A.—That he knows very little about the operation of the brake valves and distributing valve.

1033. Q.—What is the object of the supplementary reservoir of the LN equipment?

A.—To provide a high emergency brake cylinder pressure, quick recharge and a graduated release.

1034. Q.—Are all of the features of the P triple valve retained in the L valve?

A.—Yes, and in addition, quick service and a safety valve arrangement that limits the service braking ratio but is cut out when the triple valve is used in quick action.

1035. Q.—What is meant by quick service?

A.—That when the triple valve assumes service application position, each triple valve makes a local brake pipe reduction, thus positively continuing the reduction throughout the brake pipe, in a similar manner that the quick action is propagated, but in a lesser degree.

1036. Q.—What difference is there in the emergency portions of the L and P triple valves?

A.—None.

1037. Q.—What difference in the triple valve pistons?

A.—None except in size.

1038. Q.—What difference in the slide valve?

A.—It is somewhat larger and contains more ports than the valve of the P triple.

1039. Q.—What difference in the graduating valve?

A.—The graduating valve of the L triple is of a slide valve type.

1040. Q.—What piston and valve are added to the L triple valve?

A.—The by pass valve and piston.

1041. Q.—What is the piston and valve used for?

A.—To operate and admit the supplementary reservoir pressure into the auxiliary reservoir and brake cylinder during an emergency or quick action application of the brake.

1042. Q.—How is the auxiliary reservoir of the LN equipment charged?

A.—From the brake pipe through a feed groove in the piston bushing in the usual way and in addition through a port leading from the check valve case through the slide valve seat and slide valve.

1043. Q.—Why is this additional port?

A.—So that a rapid charging of the reservoir can be obtained, or rather so that both the auxiliary and supplementary can be charged in about the same time required for a P triple valve to charge an auxiliary reservoir.

1044. Q.—Why is the port opening of the additional charging port made from above the check valve?

A.—So that when a brake pipe reduction is made, the check valve will prevent any flow of pressure from the auxiliary reservoir into the brake pipe except that which can flow back through the usual feed groove.

1045. Q.—How is the supplementary reservoir charged?

A. — From the auxiliary reservoir through the triple valve slide valve and seat.

1046. Q.—At what times is the supplementary reservoir connected with the auxiliary?

A.—Only when the triple valve is in release or emergency positions.

1047. Q.—After the reservoirs are charged to a pressure equal to that in the brake pipe, what parts of the equipment are open to the atmosphere?

A.—The brake cylinder through the triple valve exhaust port, the safety valve passage and the chamber in which the emergency piston operates.

1048. Q.—How can it be determined when the charging of the reservoirs is completed?

A.—The buzzing noise made by the triple valve will cease.

1049. Q.—What causes this buzzing noise?

A.—The check valve falling to its seat due to the flow of air pressure through the charging port above the check valve.

1050. Q.—How much difference in pressure between the brake pipe and auxiliary reservoir is required to lift the check valve against the tension of the check valve spring?

A.—About 4 lbs.

1051. Q.—Explain how the triple valve operates when a service brake pipe reduction is made from the locomotive brake valve?

A.—The pressure in the brake pipe being lowered at a faster rate than at which it can flow back from the auxiliary reservoir and supplementary reservoir into the brake pipe through the feed groove, the higher auxiliary reservoir pressure moves the triple valve piston.

1052. Q.—What does the first movement of the triple valve piston result in?

A.—A closing of the feed groove, absolutely separating the auxiliary reservoir and brake pipe pressure while at the same time the graduating valve which is attached to the piston opens the service port through the slide valve.

1053. Q.—What stops the movement of the piston?

A.—Its end coming in contact with the graduating sleeve, the latter being held in place by the graduating spring.

1054. Q.—What else does the piston do on this movement?

A.—The piston engages the slide valve and brings it along to application position.

1055. Q.—What is this position called?

A.—Quick service.

1056. Q.—What port openings are made?

A.—The service port is partly opened to admit auxiliary reservoir pressure to the brake cylinder and the quick service port is opened to admit brake pipe pressure from the triple valve check case to the brake cylinder.

1057. Q.—What ports are closed by the movement of the slide valve?

A.—The brake cylinder exhaust port and the port leading from the auxiliary reservoir to the supplementary reservoir.

1058. Q.—Is the brake cylinder exhaust port closed before the service and quick service ports are opened?

A.—Yes; otherwise there would be a blow or waste of air from the triple valve exhaust port as the brake started to apply.

1059. Q.—Is the pressure in the supplementary reservoir used at this time?

A.—No, it remains bottled up.

1060. Q.—What occurs if the brake pipe reduction through leakage or be-

cause of a very short train is so rapid that the auxiliary reservoir pressure cannot expand into the brake cylinder through the partly opened service port as rapidly as the brake pipe pressure is falling?

A.—The graduating spring will be partly compressed and the valve will be moved to full service position.

1061. Q.—What is the difference between full service and quick service position?

A.—In full service position the service port is opened wide and the quick service port is entirely closed.

1062. Q.—Why is the quick service port closed?

A.—Because the brake pipe reduction is then being made at such a rapid rate as to make a quick service feature unnecessary.

1063. Q.—What is the actual object of the quick service?

A.—To continue a brake pipe reduction and insure a prompt brake application throughout a long train of cars.

*(To be continued)*

## Train Handling.

*(Continued from page 371, Dec., 1919.)*

1079. Q.—What would probably be the result of a triple valve being more sensitive to respond to brake pipe leakage than the feed valve was senstive to maintain it?

A.—Quite likely a stuck brake.

1080. Q.—Why should a stuck brake result?

A.—For the reason that brake pipe pressure is usually withdrawn at a faster rate than at which it can be replaced by the feed valve, when it is not in good condition.

1081. Q.—What is required to operate a feed valve?

A.—A differential in pressure.

1082. Q.—How much difference?

A.—Enough to overcome the tension of the supply valve piston spring and frictional resistance of the parts.

1083. Q.—About how much is this difference in pounds?

A.—From 7 to 12 lbs.

1084. Q.—Could the feed valve then be expected to operate correctly if there was less than 15 lbs. excess pressure in the main reservoir?

A.—Not under all conditions.

1085. Q.—What does this signify when the excess pressure governor top is used?

A.—That the proper amount of excess pressure is maintained and that the air gages are correct.

1086. Q.—What is excess pressure?

A.—The difference between that in the main reservoir and the brake pipe.

1087. Q.—Or in other words, what is the excess pressure at 140 lbs. main reservoir and 110 lbs. brake pipe pressure?

A.—30 lbs.

1088. Q.—Is there any difference in the length of the brake pipe exhaust at the brake valve when L triple valves are used?

A.—Yes, the exhaust is much shorter for the same brake pipe volume, as a considerable amount of the brake pipe volume is used in the brake cylinders instead of escaping to the atmosphere through the brake valve.

1089. Q.—What movement occurs if the brake pipe reduction ceases after 10, 15 or 20 lbs. pressure has been withdrawn from the brake pipe?

A.—The auxiliary reservoir continuing to expand into the brake cylinder will instantly lower the auxiliary pressure below that remaining in the brake pipe and the difference in pressure will cause the triple valve piston and graduating valve to move to lap position.

1090. Q.—To which lap position, full service lap, or quick service lap?

A.—It depends upon which position the slide valve was in when the brake pipe reduction ceased.

1091. Q.—What occurs in lap position?

A.—The flow of air to the brake cylinder, either from the auxiliary reservoir or from the brake pipe, is cut off.

1092. Q.—What occurs if the brake pipe pressure is then increased sufficiently above that in the auxiliary reservoir to overcome the frictional resistance to the movement of the piston and slide valve?

A.—The triple valve piston and the slide valves will be moved to release position, opening the brake cylinder to the atmosphere for a release of the brake and the supplementary reservoir pressure will be free to flow into the auxiliary reservoir.

1093. Q.—What will this result in if the brake pipe pressure is not maintained or if the brake valve is returned to lap position immediately after the triple valves have moved to release position?

A.—The inflow of supplementary reservoir pressure will drive the triple valve piston back to what is known as graduated release lap position.

1094. Q.—What action occurs at this time?

A.—The brake cylinder exhaust port is again closed, retaining the remainder of the brake cylinder pressure that has not had time to escape, and the flow of air from the supplementary reservoir to the auxiliary reservoir is stopped.

1095. Q.—How is this movement utilized during a release of brakes?

A.—To graduate the pressure out of the brake cylinders in any amount desired.

1096. Q.—What action results if the brake pipe pressure is maintained to a figure very much over the maximum carried when the brakes are being released, or in other words, if the brake pipe pressure increases at a faster rate than at which the auxiliary reservoir pressure increases through the inflow from the supplementary reservoir?

A.—The brake releases straight away, as with previous types of triple valves and the auxiliary reservoir is recharged from the supplementary reservoir up to within a few pounds of the original pressure.

1097. Q.—What is the advantage of this quick recharge?

A.—It requires no brake pipe pressure for a recharge of reservoirs until after the brakes have had time to release, consequently there is no drain on the brake pipe that would tend to absorb brake pipe pressure and prevent a prompt release.

1098. Q.—What will be the point of equalization between supplementary and auxiliary reservoir pressure after a 20-lb. brake pipe reduction and a release?

A.—About 105 lbs.

1099. Q.—After a 30-lb. brake pipe reduction?

A.—At about 102½ lbs.

1100. Q.—How do you estimate the pressure which the reservoir will equalize at after various brake pipe reductions?

A.—On the assumption that the reservoirs are so proportioned that a 1-lb. drop in the supplementary reservoir will increase the pressure in the auxiliary reservoir 4 lbs.

1101. Q.—Is this absolutely correct for any pressure from 0 to 100 lbs.?

A.—It may not be absolutely correct, but is close enough for all practical purposes for brake pipe reductions with standard pressures carried.

1102. Q.—What advantage has the quick recharge of the auxiliary reservoir?

A.—The brake system is at all times ready for an application.

1103. Q.—What other great advantage has this type of brake?

A.—About 4 full 20-lb. service brake applications can be made without recharge and the brake will still be more efficient than the P. M. or High-Speed brake with a 110-lb. brake pipe pressure.

1104. Q.—After 4 service applications without recharge, what would the supplementary reservoir pressure be?

A.—Approximately 90 lbs.

1105. Q.—Why would the brake be more efficient at this time than the P. M. brake with 110 lbs. in the auxiliary reservoirs?

A.—Because with an emergency application the P. M. brake would develop from 80 to 88 lbs. brake cylinder pressure which would be blown down to 60 lbs. during the stop while the L. N. brake with 90 lbs. would develop about 84 lbs. brake cylinder pressure and retain this to the stop.

1106. Q.—Explain the action of the brake if the brake valve handle is placed